Guillaume Keifer U. Arenas

Cainta, Rizal | guillaume.arenas0712@gmail.com | LinkedIn | GitHub | Portfolio | +639190008262

EDUCATION

De La Salle University *2021 – expected 2027*
Bachelor of Science and Master of Science in Manufacturing Engineering and Management
CGPA: 3.58/4.00; Candidate for Cum Laude; DOST-Merit and ERDT Scholar

EXPERIENCE

Graduate Research Assistant, Taiwan Experience Education Program *Aug 2025 – Dec 2025; May 2026 – Jul 2026*
National Chin-Yi University of Technology, Taichung, Taiwan GitHub

- Built a mechanoluminescent elastomer grid prototype and a Python feature-extraction pipeline converting video trials into light-intensity features for modeling; co-authored "Preliminary Evaluation of Mechanoluminescent Lattice-Patterned Sensing Skin for Visuotactile Contact Differentiation," published in IEEE ICCT-Pacific 2026.
- Delivered a MediaPipe-based knee-angle baseline (hip–knee–ankle landmarks), generating joint-angle time-series data adopted into an NSTC-funded project's motion-analysis workflow.
- Designed and fabricated a wrist-conforming sensor housing in Autodesk Fusion and built a low-cost automated 3-axis testing rig with a computer-vision marker-tracking pipeline to screen 4 candidate silicone elastomer blends.

Senior Officer *2024 – 2025*
Society of Manufacturing Engineers – DLSU Chapter

- Assisted in organizing events for the DLSU Department of Biomedical, Mechatronics, and Robotics Engineering, and documented 40+ students across plant visits in Osaka, Japan, curating 300+ photos into promotional material.

PROJECTS

GripVT: AI-Driven Wrist-Worn Sensor for Grip-Force Prediction (Master's Thesis) *May 2026 – Present*
Python, OpenCV, SciKit-Learn — submitted to IEEE Sensors Journal (under review) GitHub

- Developed and validated machine-learning models that predict grip force from a wrist-worn sensor to within ~6% of a wearer's maximum strength for returning wearers and ~9% for new wearers, benchmarking Ridge, XGBoost, and MLP with leakage-safe nested cross-validation on 98 synchronized sessions from 10 participants (Optuna, MLflow).
- Built the GripVT data-collection application (Python, CustomTkinter) covering calibration, live camera feed, and de-identified, crash-safe session storage; validated with a 379-test suite across 21 files (pytest).

Low-Cost Automated Visuotactile Sensor Testing Rig *May – Jun 2026*
Python, OpenCV, Ender 3 (G-code) — accepted at VTCA 2026 GitHub

- Compared 4 elastomer blends via one-way ANOVA + Tukey HSD ($p<0.001$), identifying a statistically dominant formulation and delivering a data-driven material recommendation.
- Generated 1.6M+ structured measurements per run through a real-time computer-vision pipeline (Python, OpenCV, SciPy) tracking 150+ markers/frame at ~30 FPS via Kalman filtering and Hungarian assignment, on a self-built rig (Ender 3 3D printer retrofitted with a load cell and camera, ~98% cost reduction vs. $10K–$14K commercial systems).

News Summarizer Bot *Apr 2026*
Python, Hugging Face API (BART), Web Scraping, GitHub Actions GitHub

- Built an automated news digest that scrapes 3 sources, summarizes articles through the Hugging Face API (facebook/bart-large-cnn), and delivers them via Telegram, scheduled through GitHub Actions cron jobs.
- Sustained 97.7% run uptime and a 90.8% article-summarization success rate across 88 automated runs and 1,500+ articles over 3 months of continuous operation.

Particulate Matter (PM2.5/PM10) Prediction Model — DLSU Capstone Project *Jan – Dec 2024*
Python, scikit-learn, LightGBM, OpenStreetMap GIS Data

- Compiled and cleaned 11 years (2013–2023) of PM2.5/PM10 records into a model-ready 383-sample dataset as lead of a 4-member capstone team, coordinating with 17 NCR local government units and DENR-EMB and computing road density from OpenStreetMap data.
- Developed and tuned 4 machine learning models (MLP, Random Forest, LightGBM, and a stacked RF-LightGBM ensemble) across 2 feature configurations using RandomizedSearchCV, benchmarked with 5 evaluation metrics (RMSE, MAE, MAPE, MSLE, R2), and presented findings to a faculty panel, earning a 4.0/4.0 grade.

PCB Defect Detection Research *Oct – Dec 2023*
IEEE TENCON 2024 IEEE Paper

- Optimized a YOLOv8s PCB defect detector with TensorRT, achieving 98.5% mAP50 while reducing inference time ~0.9ms and model size ~39.3% versus baseline; published in IEEE TENCON 2024.

SKILLS

Programming & Data Analysis: SQL, Python, Excel, Pandas, NumPy, Matplotlib/Seaborn, EDA, statistical analysis, C++, MATLAB
Office & Languages: Microsoft Excel, Word, PowerPoint; Google Workspace (Docs, Slides, Sheets); English (fluent), Filipino (fluent)
Machine Learning & CV: scikit-learn, TensorFlow/Keras, XGBoost/LightGBM/Ridge, Optuna, YOLO, OpenCV, MediaPipe, Hugging Face (BART)
Tools & AI: Git/GitHub, GitHub Actions, Jupyter Notebook, Claude Code, Prompt Engineering, LLM API Integration